

$$(B9) \cdot s_L d\ln W + (1-s_L) d\ln R = d\ln Y|_{K,L}$$

$$(B10) \quad d\ln W - d\ln R = g$$

Derivación $\rightarrow$ despejar $d\ln W$ y $d\ln R$

$$(B10) \cdot (1-s_L)$$

$$\hookrightarrow (1-s_L) d\ln W - (1-s_L) d\ln R = (1-s_L) g$$

$$+ (B9) \rightarrow [s_L d\ln W + \cancel{(1-s_L) d\ln R}] + [(1-s_L) d\ln W - \cancel{(1-s_L) d\ln R}] = d\ln Y|_{K,L} + (1-s_L) g$$

$$d\ln W = d\ln Y|_{K,L} + (1-s_L) g$$

$$\text{Ahora } (B10) \cdot s_L =$$

$$s_L \cdot d\ln W - s_L d\ln R = s_L g$$

Restamos de (B9):

$$[\cancel{s_L d\ln W} + (1-s_L) d\ln R] - [\cancel{s_L d\ln W} - s_L d\ln R] = d\ln Y|_{K,L} - s_L g$$

Queda:

$$d\ln R = d\ln Y|_{K,L} - s_L g$$

$\leftarrow$

Para automatización, $g < 0 \rightarrow g = -D \quad (D > 0)$

$$d\ln W = \underbrace{d\ln Y|_{K,L}}_{\text{productividad}} - \underbrace{(1-s_L) D}_{\text{desplazamiento}}, \quad d\ln R = d\ln Y|_{K,L} + s_L D$$